

The Assignment: Vigenère Cipher

Plaintext: A FATHER GIVES NOTHING BETTER THAN GOOD EDUCATION

Key: LEMON

How Vigenère works:

Ciphertext = (Plaintext letter value + Key letter value) mod 26

Step-by-step solution:

Key repeats: L E M O N L E M O N L E M O N L E M O N ...

(L=11, E=4, M=12, O=14, N=13)

Plaintext	Value	Key	Key Value	Sum	Mod 26	Cipher
A	0	L	11	11	11	L
F	5	E	4	9	9	J
A	0	M	12	12	12	M
T	19	O	14	33	7	H
H	7	N	13	20	20	U
E	4	L	11	15	15	P
R	17	E	4	21	21	V
G	6	M	12	18	18	S
I	8	O	14	22	22	W
V	21	N	13	34	8	I
E	4	L	11	15	15	P
S	18	E	4	22	22	W
N	13	M	12	25	25	Z

O	14	O	14	28	2	C
T	19	N	13	32	6	G
H	7	L	11	18	18	S
I	8	E	4	12	12	M
N	13	M	12	25	25	Z
G	6	O	14	20	20	U
B	1	N	13	14	14	O
E	4	L	11	15	15	P
T	19	E	4	23	23	X
T	19	M	12	31	5	F
E	4	O	14	18	18	S
R	17	N	13	30	4	E
T	19	L	11	30	4	E
H	7	E	4	11	11	L
A	0	M	12	12	12	M
N	13	O	14	27	1	B
G	6	N	13	19	19	T
O	14	L	11	25	25	Z
O	14	E	4	18	18	S
D	3	M	12	15	15	P
E	4	O	14	18	18	S
D	3	N	13	16	16	Q
U	20	L	11	31	5	F
C	2	E	4	6	6	G
A	0	M	12	12	12	M

T	19	O	14	33	7	H
I	8	N	13	21	21	V
O	14	L	11	25	25	Z
N	13	E	4	17	17	R

Final Ciphertext:

LJMHUPV SWIPW ZCG SMZOP XFSE ELMBT ZSPSQ FGMHVZR